CIVIC RESOLVE – AI-POWERED GRIEVANCE MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted by

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in partial fulfilment of the End Semester Examination for the course

CB23F35 –

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

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COMPUTER SCIENCE AND ENGINEERING



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BONAFIDE CERTIFICATE

Certified that this project report titled "**CIVIC RESOLVE – AI-POWERED GRIEVANCE MANAGEMENT SYSTEM**" is the bonafide work of PRASANNA. K (2116240701396), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

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DECLARATION

I, PRASANNA. K (2116240701396), hereby declare that the project report titled "CIVIC RESOLVE – AI POWERED GRIEVANCE MANAGEMENT SYSTEM", submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, pages, Deluge scripts, workflows and integrations — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai
Date: 11-11-2026

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PRASANNA. K

ABSTRACT

The management of public grievances is an important responsibility of civic administration. However, many grievance-handling processes rely on manual registers, spreadsheets, and communication between citizens and departmental officers. These practices can lead to delayed complaint resolution, difficulty tracking complaint status, duplication of records, limited transparency, and challenges in monitoring officer performance. Citizens may also face difficulties obtaining timely updates about their complaints. To address these challenges, this project proposes Civic Resolve, a centralized grievance management system developed using the low-code platform Zoho Creator.

The application is developed using Zoho Creator and includes modules for Citizen Management, Department Management, Officer Management, Complaint Management, Complaint Activity Log, Resolution Evidence, and Closure Review. It provides customized reports and dashboards to monitor complaints, officer assignments, and resolution progress. Deluge scripting and workflow automation simplify important operations such as automatic officer assignment, SLA deadline calculation, complaint status updates, activity tracking, and notifications. The Closure Review module allows reviewers to approve or reject complaint closure requests and automatically updates the corresponding complaint status. Role-based access control provides appropriate permissions for administrators, citizens, officers, and reviewers, ensuring secure and organized access to the system.

The system is tested through functional scenarios covering complaint submission, officer assignment, status updates, activity logging, and closure approval and rejection. These tests help verify record consistency and the expected execution of automated workflows. The application provides a centralized approach to grievance administration by reducing repetitive manual activities, minimizing duplicate data entry, improving complaint traceability, and supporting more efficient monitoring of resolution timelines. Overall, Civic Resolve demonstrates how low-code development and workflow automation can improve the accessibility, transparency, and efficiency of civic grievance management.

Keywords: *Low-Code Development; Zoho Creator; Deluge Scripting; Workflow Automation; Grievance Management; Role-Based Access Control.*

PROJECT AT A GLANCE

Attribute	Details
Project Title	Civic Resolve – AI-Powered Civic Grievance Management System
Application Domain	Civic Administration / Grievance Management
Student Name & Register No.	Prasanna. K – 2116240701396
Zoho Creator Edition	Trial Edition (Trial currently active)
Application Name (Link Name)	Civic Resolve Grievance System (CRG)
Published / Live URL	https://prasannak2024cserajal744.zohocreatorportal.in
Demo Credentials (per role)	System Administrator, Department Head, Grievance Officer, Compliance Reviewer, Citizen, Citizen Advocate
Number of Forms / Subforms	9 / 4
Number of Reports (by type)	List: 9; Kanban: 7; Calendar: 8; Pivot/Chart: 0; Spreadsheet: 9; Map: 1; Timeline: 8
Number of Pages / Dashboards	1 Dashboard page
Number of Workflows	Form: 23; Schedule: 4; Approval: 1; Blueprint: 3; Batch: 3;
Deluge Functions / Custom Actions	6 custom Deluge functions
Integrations	Zoho CRM (Contacts and Cases), Zoho Zia OCR, Zoho Zia Predictive Analytics, automated email notifications
Roles & Profiles Defined	5 Roles and 11 Profiles
Environments Used	Development → Stage → Production
Date of Production Deployment	08-11-2026

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
ER	Entity–Relationship
JSON	JavaScript Object Notation
LCAP	Low-Code Application Platform
OAuth	Open Authorization
RBAC	Role-Based Access Control
REST	Representational State Transfer
SDLC	Software Development Life Cycle
SSO	Single Sign-On
UAT	User Acceptance Testing
UI	User Interface
Zia	Zoho's AI assistant
CRM	Customer Relationship Management
AI	Artificial Intelligence
ID	Identifier
URL	Uniform Resource Locator

CHAPTER 1
INTRODUCTION

1.1 Background

Civic Resolve belongs to the Civic Administration and Public Grievance Management domain and is designed for municipal and local government organisations responsible for addressing public complaints related to sanitation, road maintenance, water supply, street lighting, and other civic services. The system is intended to support organisations handling complaints from multiple citizens across different departments, requiring coordination between administrators, departmental officers, and reviewers. It addresses the complete grievance management process, including complaint registration, department assignment, officer allocation, status tracking, resolution evidence submission, and closure review. By providing a centralized digital platform, the project aims to reduce manual recordkeeping, improve transparency, monitor complaint progress, and support timely resolution.

1.2 Problem Statement

Traditional civic grievance management processes often depend on manual registers, spreadsheets, and disconnected communication methods to handle public complaints. These methods create difficulties in managing complaint records, assigning complaints to the appropriate officers, monitoring resolution progress, and ensuring timely closure. The absence of a centralized system reduces transparency, delays decision-making, and makes it difficult for administrators to monitor departmental performance. Therefore, Civic Resolve aims to provide a centralized, automated, and secure platform using Zoho Creator to streamline complaint registration, officer assignment, status tracking, resolution evidence management, and closure approval.

- **Data Duplication:** Complaint details are maintained across multiple registers and spreadsheets, increasing the possibility of duplicate and inconsistent records.
- **Delayed Processing:** Manual complaint assignment and approval procedures can delay complaint resolution.
- **Lack of Real**-Time Tracking: Citizens and administrators have limited visibility into complaint status and resolution progress.
- **Manual Monitoring:** Tracking officer activities, SLA deadlines, and pending complaints requires additional administrative effort.

1.3 Objectives

- To design a normalised data model and forms for citizen, department, officer, complaint, activity log, resolution evidence, and closure review management.
- To automate complaint registration, officer assignment, SLA calculation, status updates, and closure approval using Deluge scripting and workflows.
- To provide role-based dashboards and reports for monitoring complaints, officer activities, pending cases, and resolution progress.
- To integrate the application with Zoho service for data management, notifications, and workflow automation.
- To deploy the application securely to a production environment with governance controls.

1.4 Scope of the Project

The project covers the development of a web-based civic grievance management application using Zoho Creator. The in-scope features include citizen, department, and officer management; complaint registration and assignment; SLA monitoring; complaint status tracking; activity logging; resolution evidence submission; closure review; automated notifications; and role-based reports and dashboards. The system supports four user roles: Administrator, Citizen, Officer, and Reviewer, with appropriate access permissions. It focuses on managing the grievance process from complaint registration to final closure through a centralized platform. The project does not include automatic complaint registration through external government portals, real-time GPS-based tracking, AI-powered complaint classification, or direct integration with municipal government databases. Native mobile application development is also outside the current scope.

1.5 Justification for a Low-Code Approach

Zoho Creator is selected for developing Civic Resolve because its low-code approach enables faster application development with minimal manual coding. Its drag-and-drop interface simplifies the creation of forms, reports, dashboards, and workflows, while Deluge scripting supports customized business logic and automation. The platform provides built-in cloud hosting, web and mobile accessibility, data management, and role-based access controls, reducing infrastructure and maintenance requirements. It is also suitable for future modifications and maintenance by developers and citizen developers. These features make Zoho Creator a practical and cost-effective choice for developing a centralized civic grievance management application.

1.6 Organisation of the Report

Chapter 2 reviews the existing system and related low-code literature. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design. Chapter 5 details the implementation on Zoho Creator. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, performance optimisation, security and governance. Chapter 8 presents results and discussion, and Chapter 9 concludes the report with future enhancements.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In the existing civic grievance management process, complaints are generally received from citizens through direct visits, phone calls, or written applications. The details of each complaint are manually recorded in registers or spreadsheets by administrative staff. The complaints are then reviewed and forwarded to the respective departments for further action. Department officials manually assign complaints to available officers, who investigate the issue and carry out the necessary resolution activities. Citizens may need to contact the department to obtain updates about their complaint status. Once the issue is resolved, the officer informs the concerned department, and the complaint is manually marked as closed after verification. However, this process involves repeated manual work, delays in communication, difficulty tracking complaint progress, and limited visibility into officer activities and pending complaints. Maintaining historical records and preparing performance reports also require additional administrative effort.

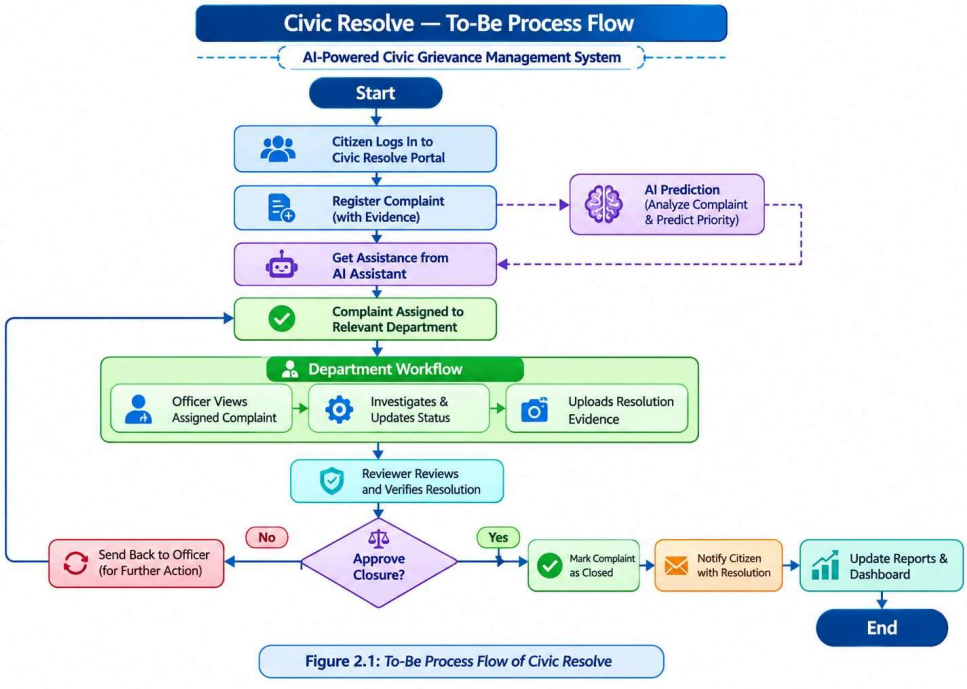


Figure 2.1: As-Is Process Flow of the Civic Resolve Grievance Management System

2.2 Limitations of the Existing System

- **Manual Data Management:** Maintaining complaint records using registers and spreadsheets increases the risk of data duplication, errors, and inconsistent information.
- **Delayed Complaint Resolution:** Manual department assignment, officer allocation, and approval processes can cause delays in handling and resolving complaints.
- **Lack of Transparency and Monitoring: Citizens** and administrators have limited visibility into complaint status, officer activities, pending cases, and resolution progress.

2.3 Low-Code versus Traditional Development

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	Approximately 8–10 weeks	Approximately 2–3 weeks
Skills required	HTML, CSS, JavaScript, backend programming, database management	Zoho Creator, Deluge scripting, basic application development
Hosting & infrastructure	Requires separate server setup, database, and hosting configuration	Built-in cloud hosting and database services
Mobile application	Requires additional responsive development or mobile application development	Supports mobile access through Zoho Creator
Security & access control	Requires manual implementation of authentication, authorization, and security	Built-in user management, permissions, and role-based access
Maintenance & change	Requires code modifications, testing, and redeployment	Easier modifications using visual builders and workflows
Customisation limits	Provides greater flexibility and complete control over application features	Customisation depends on platform capabilities and Deluge scripting
Cost	Higher development and infrastructure costs	Lower initial development and infrastructure costs

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

2.4 Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	Bucaioni, Cicchetti & Ciccozzi (2022)	Modelling in Low-Code Development: A Multi-Vocal Systematic Review	Identifies visual modelling, abstraction, automation, and improved development productivity as important low-code characteristics.	Supports the use of Zoho Creator's visual development tools for building Civic Resolve.
2	Bock & Frank (2021)	Low-Code Platform	Explains how low-code platforms support software productivity, business-IT alignment, and citizen development.	Justifies using a low-code platform to develop and maintain the grievance management application.
3	Pinho, Aguiar & Amaral (2023)	What About the Usability in Low-Code Platforms? A Systematic Literature Review	Highlights usability as an important factor for users with limited programming experience.	Supports the development of simple forms and user-friendly interfaces for different user roles.
4	Zoho Corporation (n.d.)	Zoho Creator – Applications and Deluge Documentation	Explains forms, reports, pages, workflows, and Deluge scripting for application development and automation	Provides the technical foundation for complaint registration, reports, dashboards, and automated workflows.
5	CivicPlus – SeeClickFix (n.d.)	SeeClickFix 311 CRM – Request Management	Provides features for service-request submission, assignment, status updates, request tracking, and staff management.	Serves as a reference for designing complaint assignment, tracking, and resolution processes in Civic Resolve.

Table 2.2: Summary of Literature Survey

2.5 Emerging Trends in Low-Code Development

Low-code development is evolving through emerging trends such as AI-assisted application building, citizen development, hyper-automation, and composable applications. AI-assisted tools help developers create application components and workflows more efficiently, while citizen development enables users with basic technical knowledge to build and maintain applications. Hyper-automation combines workflows, scripting, and automated notifications to reduce repetitive manual tasks. Composable applications allow different modules and services to work together as a flexible system. These trends influenced the design of Civic Resolve by encouraging the use of Zoho Creator's visual application builder, Deluge scripting, automated workflows, reusable forms, reports, and role-based access. This approach simplifies complaint management, improves process efficiency, and supports future application enhancements.

2.6 Proposed System

The proposed Civic Resolve system is a centralized web-based grievance management application developed using Zoho Creator. It provides modules for managing citizens, departments, officers, complaints, activity logs, resolution evidence, and closure reviews. The system automates complaint registration, officer assignment, SLA deadline calculation, status updates, activity tracking, and closure approval using Deluge scripting and workflows. Customized reports and dashboards provide better visibility into complaint progress and officer activities. Role-based access ensures that users can access only the features permitted for their roles. The proposed system overcomes the limitations of the existing process by reducing manual data entry, minimizing duplicate records, speeding up complaint assignment and approval, improving complaint tracking, and providing organized monitoring of pending and resolved cases. Overall, it aims to improve the efficiency, transparency, and accountability of civic grievance management.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Administrator	Manages and supervises the overall grievance management system.	Manage citizens, departments, and officers; configure application settings; manage user access; view all complaint records and reports.
Manager (Reviewer)	Supervises complaint resolution and reviews closure requests.	Review complaints, approve or reject closure requests, provide review remarks, and monitor resolution progress.
Staff / Operator (Officer)	Handles complaints assigned by the administration.	View assigned complaints, update complaint status, record actions taken, and submit resolution evidence
Customer / Portal User (Citizen)	A citizen who registers complaints and monitors their progress.	Submit complaints, provide complaint details, view complaint status, and track resolution progress.

Table 3.1: Stakeholders and User Roles

3.2 Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall allow citizens to register new complaints by providing mandatory details such as complaint title, description, department, and supporting information.	Citizen	High
FR-02	The system shall automatically calculate the complaint submission date and SLA deadline upon submission.	Administrator, Citizen	High
FR-03	The system shall send email notifications to assigned officers when a complaint is assigned or its status is updated.	Officer, Administrator	Medium
FR-04	The system shall route complaint closure requests to the assigned reviewer for approval or rejection.	Officer, Reviewer	High
FR-05	The system shall display dashboards and reports showing total complaints, pending complaints, resolved complaints, and officer activities.	Administrator, Reviewer	Medium

FR ID	Requirement	Role(s)	Priority
FR-06	The system shall maintain citizen, department, and officer details in centralized records for complaint management.	Administrator	High
FR-07	The system shall allow officers to update complaint status and record remarks about actions taken.	Officer	High
FR-08	The system shall maintain an activity log containing complaint status changes, assigned officers, departments, and timestamps.	Administrator, Officer, Reviewer	High
FR-09	The system shall allow officers to upload resolution evidence for assigned complaints.	Officer	Medium
FR-10	The system shall allow reviewers to approve or reject closure requests and automatically update the corresponding complaint status.	Reviewer	High
FR-11	The system shall allow citizens to view and track the status of their submitted complaints.	Citizen	High
FR-12	The system shall restrict access to application modules and records according to assigned user roles and permissions.	All Roles	High

Table 3.2: Functional Requirements

3.3 Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	Reports containing up to 1,000 records shall load within 5 seconds under normal operating conditions.
NFR-02	Security	Users shall access only the data and application features permitted by their assigned roles.
NFR-03	Usability	The application shall be accessible through web browsers and the Zoho Creator mobile application with a user-friendly interface.
NFR-04	Availability	The application shall be accessible through the internet whenever the Zoho Creator service is available.
NFR-05	Maintainability	The application shall support easy updates to forms, reports, workflows, and Deluge scripts whenever business requirements change.
NFR-06	Auditability	Complaint creation, status changes, officer assignments, and review activities shall be traceable through activity logs.
NFR-07	Reliability	The system shall maintain accurate complaint records and ensure that submitted information is stored consistently.

NFR ID	Category	Requirement
NFR-08	Scalability	The application shall support an increasing number of citizens, complaints, departments, and officers within the limits of the selected Zoho Creator plan.

Table 3.3: Non-Functional Requirements

3.4 Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator Education Edition
Scripting language	Deluge (Data Enriched Language for the Universal Grid Environment)
Browser	Google Chrome / Microsoft Edge / Mozilla Firefox (latest version)
API testing tool	Postman
Other Zoho services	Zoho Creator built-in services for forms, reports, workflows, notifications, and data management.
Mobile	Zoho Creator mobile app (Android / iOS) if used
Client hardware	Intel Core i3/i5 or above, 8 GB RAM, internet connection

Table 3.4: Hardware and Software Requirements

3.5 Use Case Model

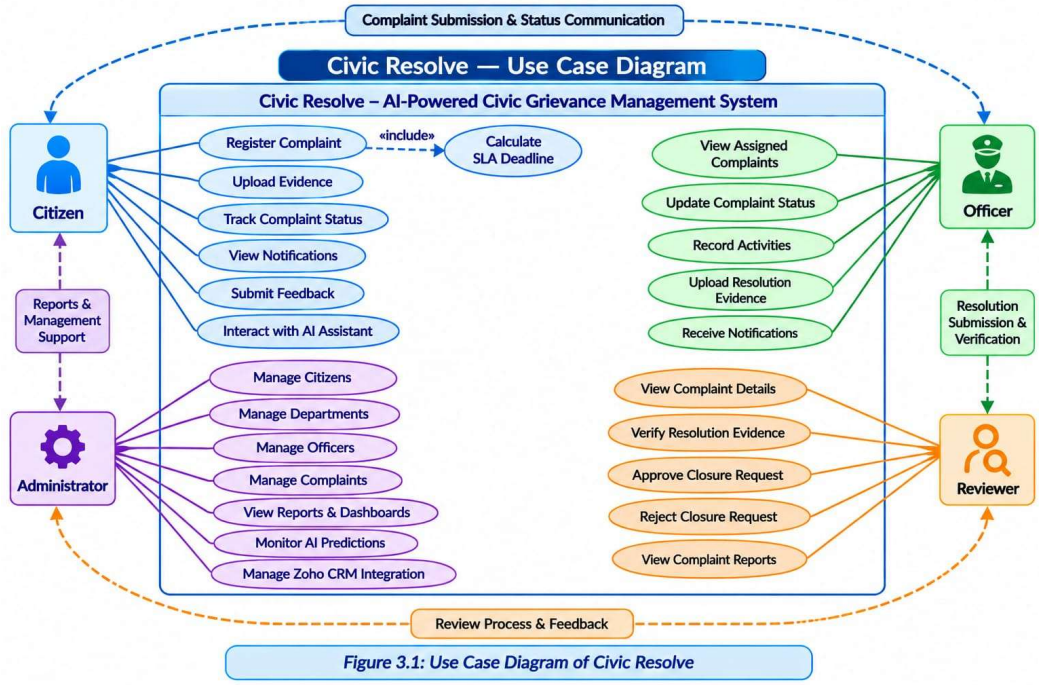


Figure 3.1: Use Case Diagram of Civic Resolve – AI-Powered Civic Grievance Management System

Use Case ID	UC-01: Register Complaint
Use Case Name	Register Complaint
Actor(s)	Citizen
Pre-condition	Citizen must have access to the application and provide the required complaint details.
Main Flow	1. Citizen opens the Complaint form. 2. Enters complaint details. 3. Selects the concerned department. 4. Uploads supporting evidence if available. 5. Submits the complaint. 6. System saves the complaint and generates a complaint record.
Alternate Flow	If mandatory fields are missing, the system displays a validation message and requests the citizen to enter the required details.
Post-condition	Complaint is successfully registered and stored in the system with a submission date and SLA deadline.
Related FR	FR-01, FR-02

Table 3.5: Use Case Description – [Register Complaint]

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

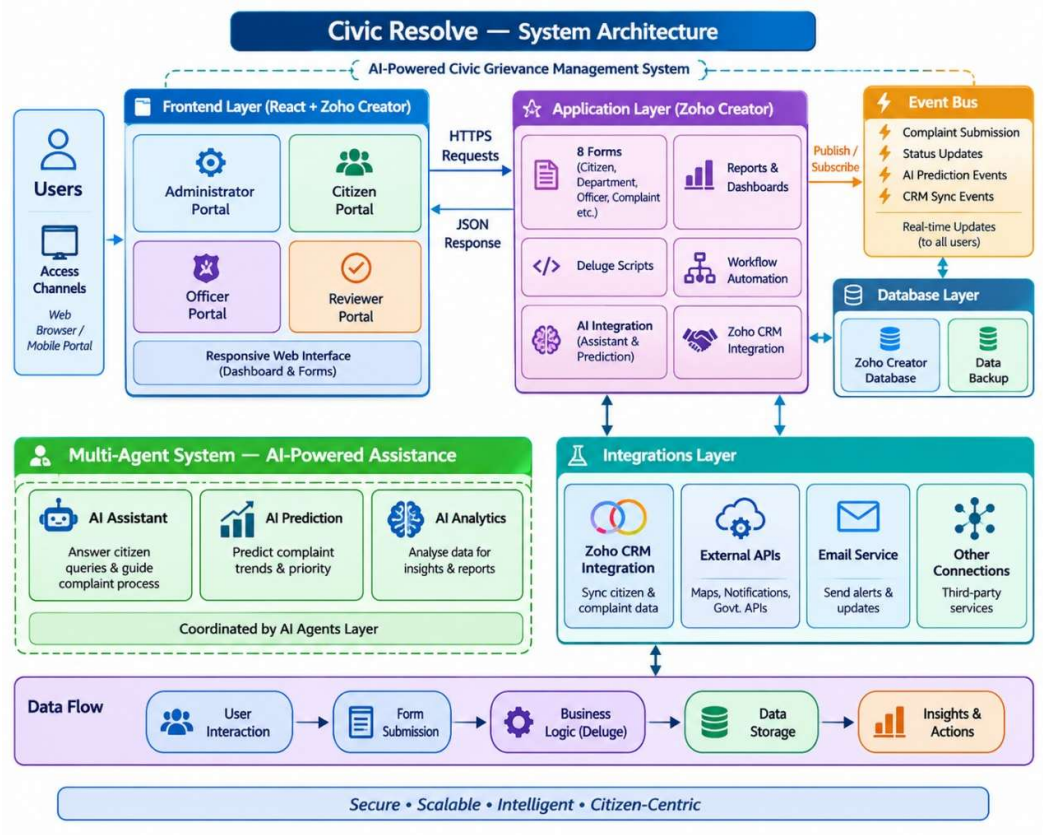


Figure 4.1: System Architecture of Civic Resolve – AI-Powered Civic Grievance Management System

Civic Resolve follows a layered architecture developed using Zoho Creator. The user layer consists of Administrators, Citizens, Officers, and Reviewers who access the application through web browsers or the Zoho Creator mobile app. The application layer contains forms, reports, dashboards, and pages for managing complaints and related records. The business logic layer uses Deluge scripting and workflows to automate officer assignment, SLA calculation, status updates, activity logging, notifications, and closure approvals. The data layer stores citizen, department, officer, complaint, activity log, resolution evidence, and closure review records in Zoho Creator's built-in database. The integration layer uses Zoho Creator's built-in services for data management and notifications.

4.2 Data Model

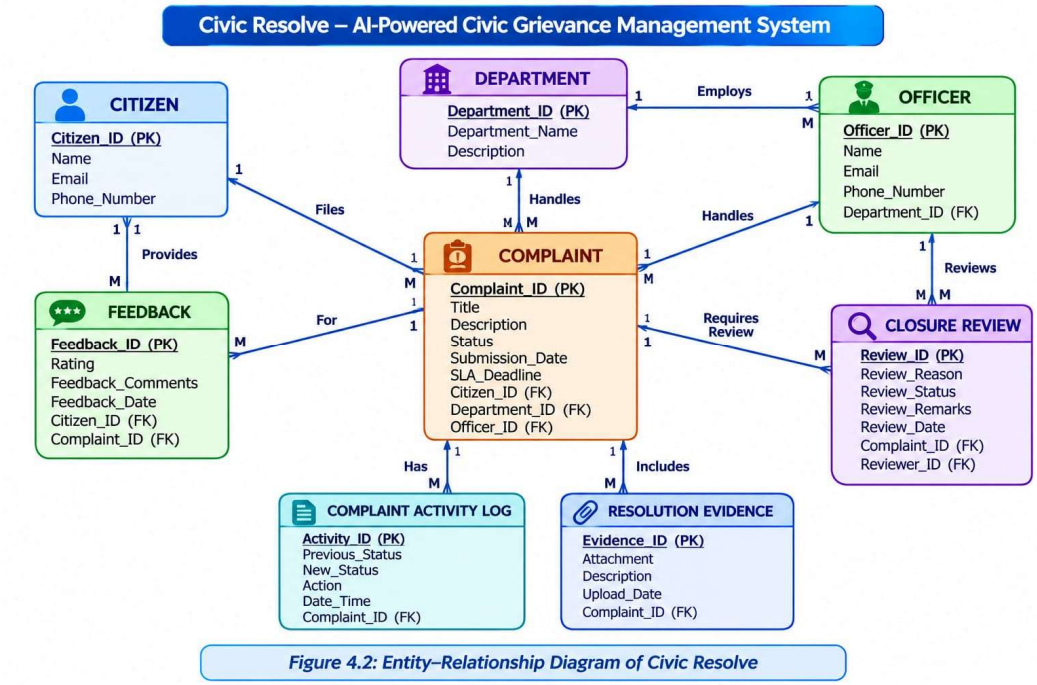


Figure 4.2: Entity–Relationship Diagram of Civic Resolve – AI-Powered Civic Grievance Management System

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Citizen	Citizen_ID, Name, Email, Phone_Number	Complaint, Feedback	One-to-Many	Lookup in Complaint and Feedback
Department	Department_ID, Department_Name, Description	Officer, Complaint	One-to-Many	Lookup in Officer and Complaint
Officer	Officer_ID, Name, Email, Phone_Number	Department, Complaint	Many-to-One / One-to-Many	Lookup fields in Officer and Complaint
Complaint	Complaint_ID, Title, Description, Status, Submission_Date, SLA_Deadline	Citizen, Department, Officer, Complaint Activity Log, Resolution Evidence, Closure Review, Feedback	Many-to-One / One-to-Many	Lookup fields in Complaint and related forms

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Complaint Activity Log	Activity_ID, Previous_Status, New_Status, Action, Date_Time	Complaint	Many-to-One	Lookup in Complaint Activity Log
Resolution Evidence	Evidence_ID, Attachment, Description, Upload_Date	Complaint	Many-to-One	Lookup in Resolution Evidence
Closure Review	Review_ID, Review_Reason, Review_Status, Review_Remarks, Review_Date	Complaint, Reviewer (Officer)	Many-to-One	Lookup fields in Closure Review
Feedback	Feedback_ID, Rating, Feedback_Comments, Feedback_Date	Citizen, Complain	Many-to-One	Lookup fields in Feedback

Table 4.1: Entities and Relationships

4.3 Form Design

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Citizen ID	Citizen_ID	Auto Number	—	Prefix CIT-	System generated
Name	Name	Name	Yes	—	Citizen's full name
Email	Email	Email	Yes	No duplicate values	Citizen's email address
Phone Number	Phone_Number	Phone	Yes	Valid phone number	Citizen's contact number

Table 4.2: Field Specification – Citizen Form

4.4 Process and Workflow Design

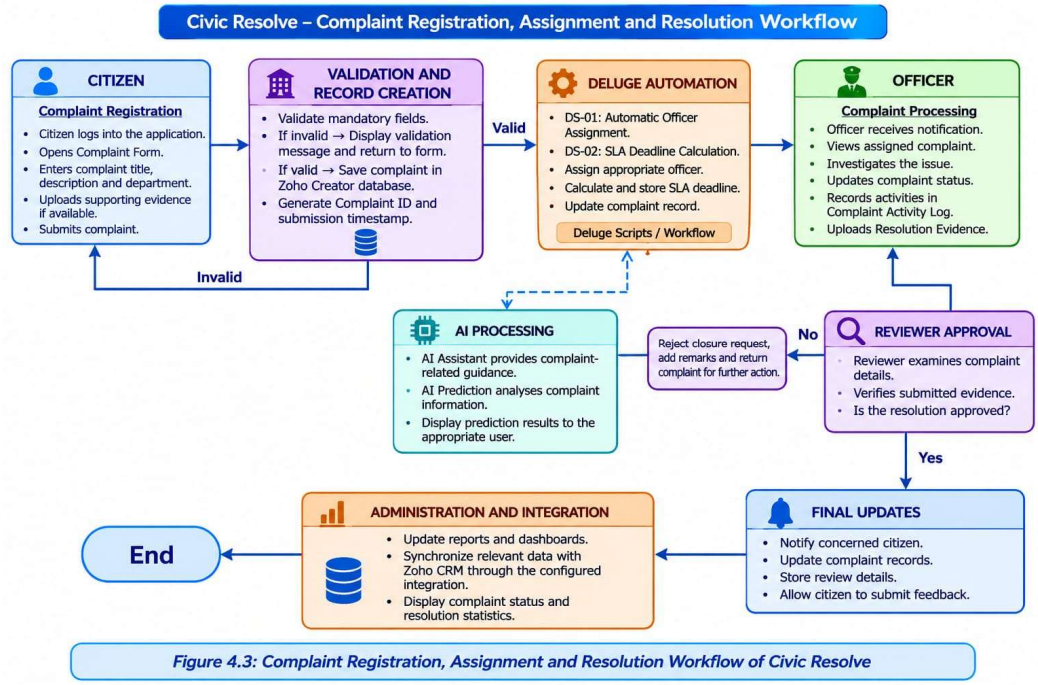


Figure 4.3: Complaint Registration, Assignment and Resolution Workflow of Civic Resolve

Figure 4.3: Complaint Registration, Assignment and Resolution Workflow of Civic Resolve

4.5 User Interface and Navigation Design

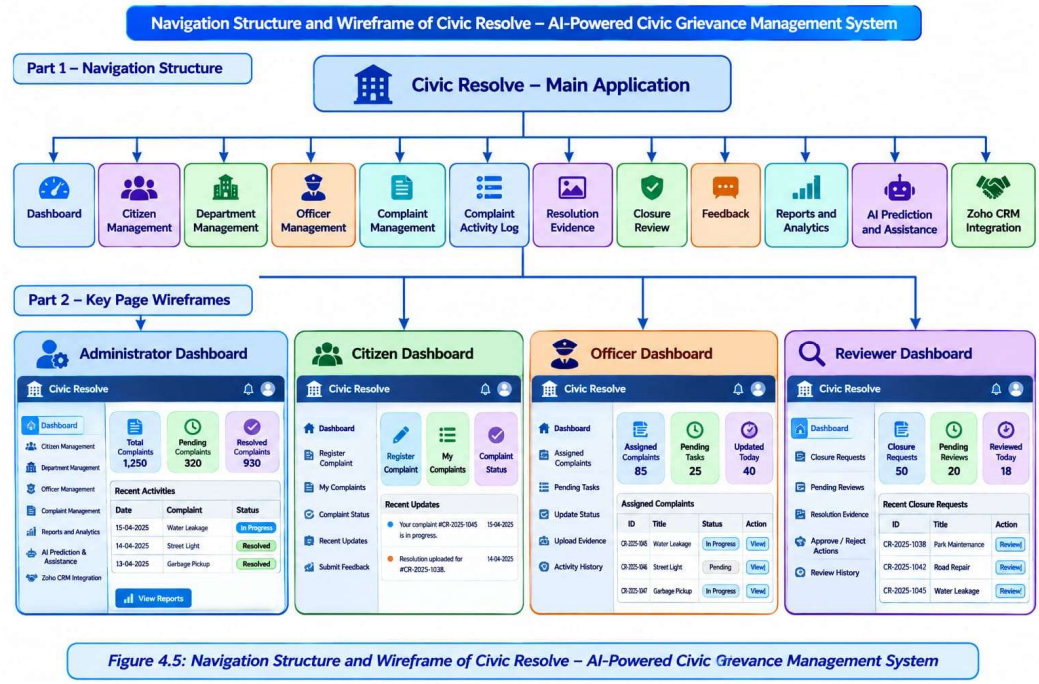


Figure 4.5: Navigation Structure and Wireframe of Civic Resolve – AI-Powered Civic Grievance Management System

Figure 4.4: Navigation Structure and Wireframe of Civic Resolve

4.6 Access Control Design

Component	Administrator	Manager	Staff	Portal User
Form: Citizen	A E V D	V	V	V (own)
Form: Department	A E V D	V	V	V
Form: Officer	A E V D	V	V (own)	—
Form: Complaint	A E V D	E V	E V (assigned)	A V (own)
Form: Complaint Activity Log	A V	V	AV	A V (own)
Form: Resolution Evidence	A E V D	V	A E V (assigned)	V (own)
Form: Closure Review	A E V D	AEV	V	V (own)
Form: Feedback	A V	V	—	A V (own)
Report: All Complaints	V	V	V (assigned)	V (own)
Page: Admin Dashboard	V	—	—	—
Page: Manager Dashboard	V	V	—	—
Page: Officer Dashboard	V	—	V	—
Page: Citizen Dashboard	V	—	—	V (own)

Table 4.3: Role–Permission Matrix

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

Civic Resolve was developed using the Zoho Creator low-code platform. The application was created from scratch using the Zoho Creator application builder. The workspace was configured to organize the application into modules for Citizen, Department, Officer, Complaint, Complaint Activity Log, Resolution Evidence, Closure Review, and Feedback management. These modules provide a centralized structure for handling civic grievances from complaint registration to final closure and feedback collection. Forms were created using the drag-and-drop form builder, while reports and dashboards were configured to manage and monitor application data. Deluge scripting and workflows were used to automate operations such as officer assignment, SLA deadline calculation, complaint status updates, activity tracking, notifications, and closure review. The application navigation was organized to provide access to the required modules based on user roles. A consistent interface and application theme were maintained to support easy navigation and improve usability.

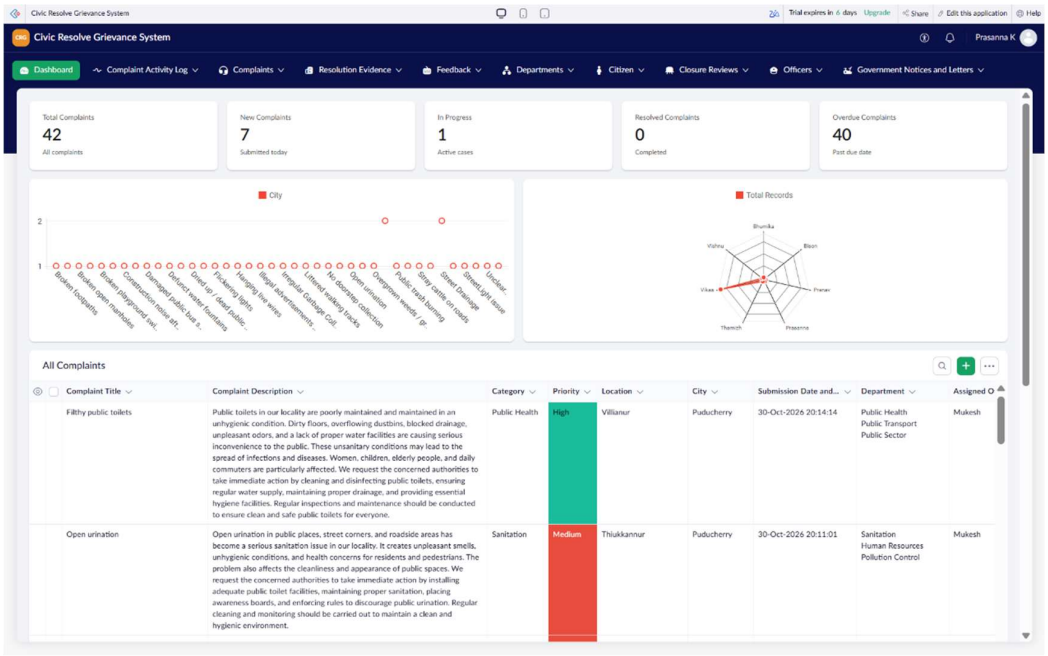


Figure 5.1: Application Home and Navigation Sections of Civic Resolve

5.2 Forms and Data Capture

The Civic Resolve application consists of eight forms developed using the Zoho Creator Form Builder. Each form is designed to capture and manage specific information required for the grievance management process. The Citizen Form stores citizen details such as name, email address, and phone number. The Department Form maintains department information, while the Officer Form stores officer details and connects officers to their respective departments through lookup fields. The Complaint Form captures complaint title, description, citizen details, department, assigned officer, submission date, SLA deadline, and complaint status.

The Complaint Activity Log Form records complaint-related activities, status changes, remarks, and timestamps. The Resolution Evidence Form allows officers to upload supporting documents and descriptions as proof of completed work. The Closure Review Form manages complaint closure requests, review status, and reviewer remarks. The Feedback Form collects citizen ratings and comments about the complaint resolution process.

Lookup fields are used to establish relationships between the forms and their associated records. Mandatory field settings and field-level validations help maintain accurate and complete data. The forms use individual records connected through lookup fields rather than subforms. Form properties provide submission and reset controls, along with confirmation messages after successful submission. These configurations support structured data capture and the complete grievance management process, from complaint registration to final closure and feedback collection.

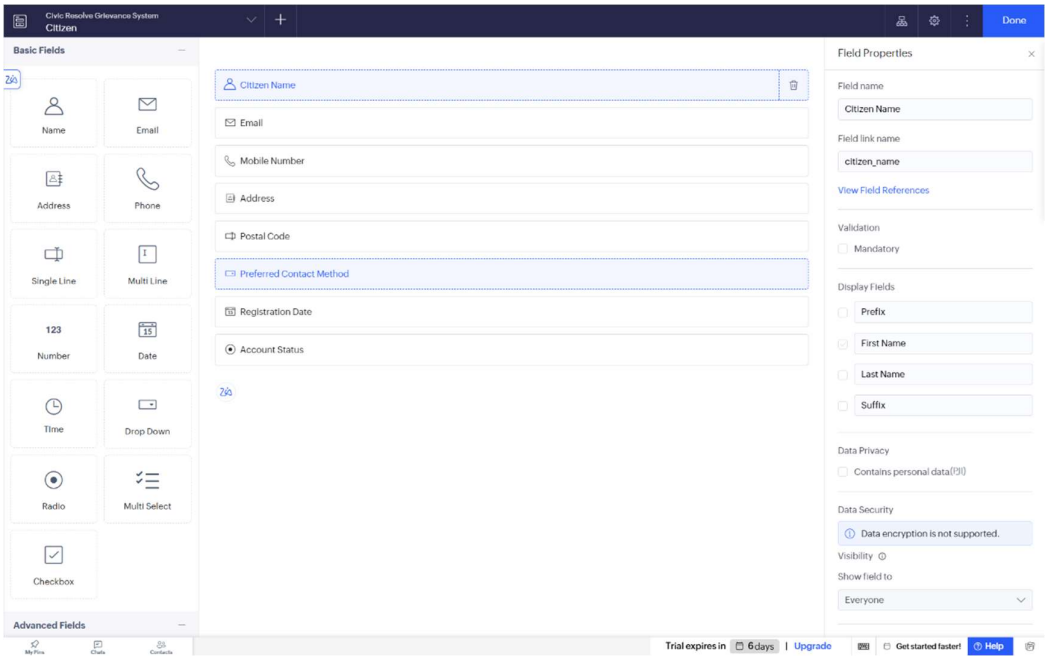


Figure 5.2: Citizen Form in the Form Builder

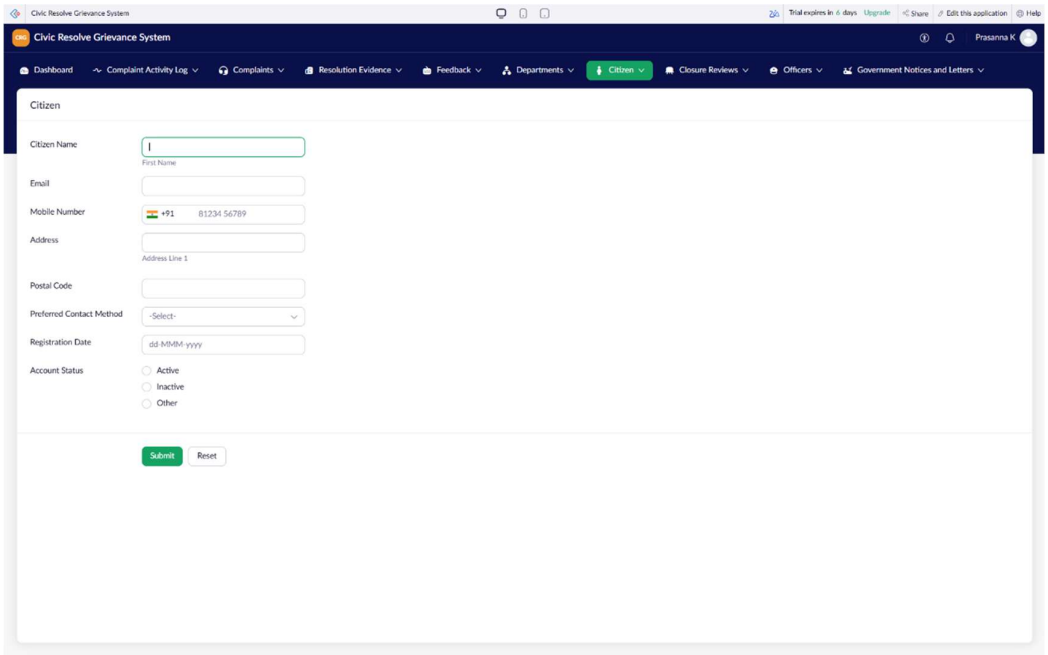


Figure 5.3: Citizen Form in Live Mode

5.3 Reports

Report Name	Type	Source Form	Filter / Grouping / Sorting	Custom Actions	Roles
All Complaints	List	Complaint	Sort by Submission Date (Descending)	View / Edit	Administrator, Reviewer
Complaint Status Report	Pivot / Chart	Complaint	Group by Complaint Status	View Details	Administrator, Reviewer
Officer-wise Complaints	List	Complaint	Group by Assigned Officer	View Details	Administrator, Reviewer, Officer
Pending Complaints	List	Complaint	Filter: Status is not Closed; Sort by SLA Deadline	View / Edit	Administrator, Reviewer, Officer
Complaint Activity Report	List	Complaint Activity Log	Sort by Date and Time (Descending)	View Details	Administrator, Reviewer
Resolution Evidence Report	List	Resolution Evidence	Group by Review Status	View Attachment	Administrator, Reviewer, Officer
Closure Review Report]	List	Closure Review	Group by Review Status	View Review Details	Administrator, Reviewer
Citizen Feedback Report	List / Chart	Feedback	Group by Rating	View Feedback	Administrator, Reviewer

Table 5.1: Reports Implemented

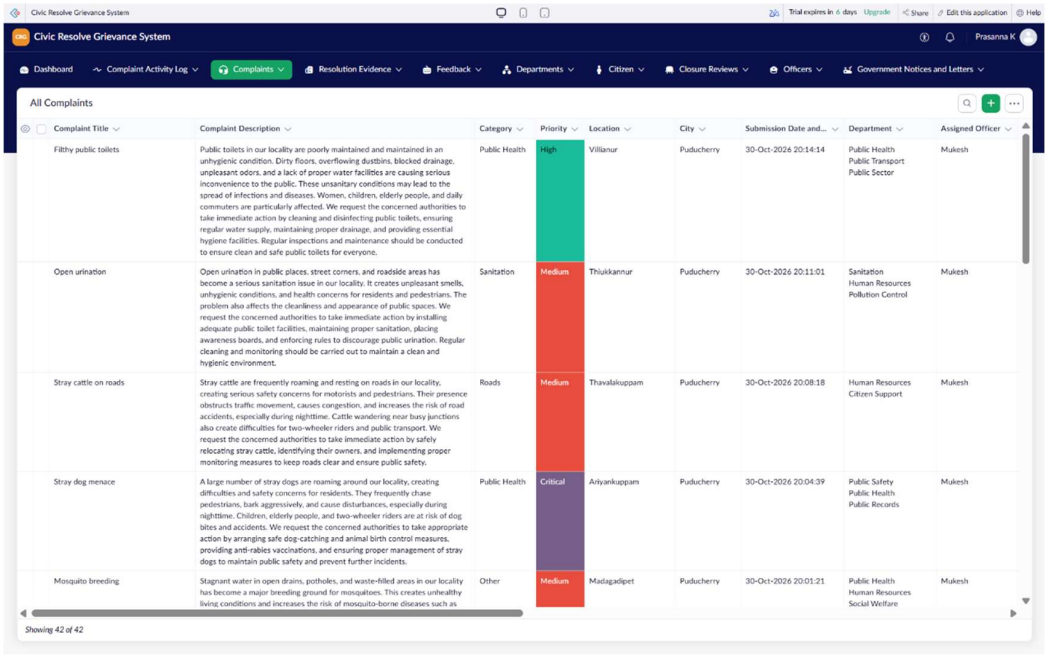


Figure 5.4: All Complaints Report

5.4 Pages and Dashboards

The Civic Resolve application uses pages and dashboards to provide role-based access to important grievance management information. The Administrator Dashboard is designed for administrators to monitor the overall application and manage citizen, department, officer, and complaint records. It provides access to complaint reports, pending cases, resolved complaints, and officer-related information. The Manager Dashboard (Reviewer) supports complaint monitoring and closure verification by providing access to complaint details, closure review records, and resolution progress. The Officer Dashboard helps officers manage their assigned complaints, update complaint statuses, record activities, and submit resolution evidence. The Citizen Dashboard allows citizens to access their submitted complaints, track complaint status, view progress, and submit feedback after resolution.

The dashboards use Zoho Creator page components such as embedded reports, forms, buttons, and summary information to organize application data. Embedded reports provide access to complaint records and related activities, while forms support data entry and updates. Navigation buttons help users move between the available modules. Role-based access permissions ensure that each user can access only the pages and information relevant to their responsibilities. These pages provide a centralized

interface for monitoring complaints, managing activities, reviewing resolutions, and collecting citizen feedback.

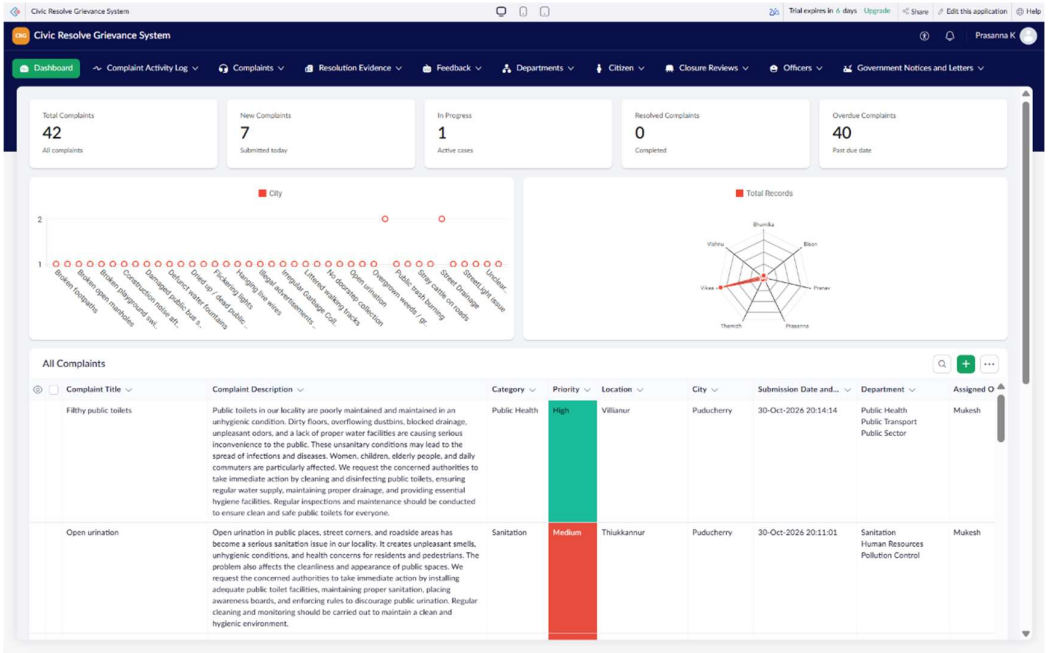


Figure 5.5: Administrator Dashboard of Civic Resolve

5.5 Deluge Scripting

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-01	Complaint Form → On Success	Automatically assign an appropriate officer to a registered complaint.	Fetch records, criteria, conditional statements, record update
DS-02	Complaint Form → On Success	Calculate the SLA deadline based on the complaint submission date.	Date functions, variables, date calculation
DS-03	Complaint Form / Workflow → On Success	Update complaint status and maintain a record of complaint activities.	If-else, record fetching, record insertion, field updates
DS-04	Complaint Status Workflow	Send notifications when complaints are assigned or their status changes.	Sendmail, string concatenation, conditional statements
DS-05	Resolution Evidence / Workflow	Maintain resolution evidence associated with the complaint.	Lookup, record fetching, record insertion
DS-06	Closure Review Form → On Success	Process closure review decisions and update the related complaint status.	Lookup, if-else, criteria, record update

Table 5.2: Deluge Scripts Implemented

Code 5.1: Automatic SLA Deadline Calculation

```
// DS-02 | Complaint Form | On Success
// Purpose: Calculate the SLA deadline

// Use the configured SLA duration for the complaint.
slaDays = 3;
input.SLA_Deadline = input.Submission_Date.addDay(slaDays);
```

Code 5.2: Closure Review Status Update

```
// DS-06 | Closure Review Form | On Success
// Purpose: Update the related complaint status

if(input.review_status == "Approved")
{
    complaintRecord = complaint[ID == input.complaint];

    if(complaintRecord.count() > 0)
    {
        complaintRecord.status = "Closed";
    }
}
else if(input.review_status == "Rejected")
{
    complaintRecord = complaint[ID == input.complaint];

    if(complaintRecord.count() > 0)
    {
        complaintRecord.status = "Under Review";
    }
}
```

The Deluge scripts in Civic Resolve automate important operations involved in complaint management. The automatic officer assignment script identifies the appropriate department and assigns a suitable officer to the registered complaint, reducing manual assignment work. The SLA calculation script calculates the complaint resolution deadline using the submission date and configured SLA period, helping administrators monitor pending complaints. The complaint status update and activity logging script maintains the current complaint status and records activities such as status changes and officer actions, providing a history of complaint progress. The notification script sends updates to concerned users when complaints are assigned or their status changes. The resolution evidence script associates uploaded evidence with the corresponding complaint, allowing reviewers to verify the completed work. The closure review script checks the reviewer's decision and updates the complaint status to Closed when approved or Under Review when rejected.

These scripts reduce repetitive manual operations and maintain consistency across the grievance management process. The expected results include automatic deadline calculation, organized officer assignment, updated complaint records, activity tracking, timely notifications, and controlled complaint closure. The results should be described as successfully observed only for scripts that were tested and executed correctly in Zoho Creator.

5.6 Workflows and Automation

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Ref.
Automatic Officer Assignment	Form Workflow	Complaint → On Success	When a new complaint is submitted	Automatically assign an appropriate officer to the complaint	DS-01
SLA Deadline Calculation	Form Workflow	Complaint → On Success	When a complaint is registered	Calculate and store the SLA deadline	DS-02
Complaint Status Update	Form Workflow	Complaint status update	When the complaint status changes	Update complaint details and record the activity in the Activity Log	DS-03
Complaint Notification	Form Workflow	Complaint assignment / status update	When an officer is assigned or complaint status changes	Send email notifications to concerned users	DS-04
Closure Review Processing	Form Workflow	Closure Review → On Success	When review status is Approved or Rejected	Update the related complaint status to Closed or Under Review	DS-05
Resolution Evidence Management	Form Workflow	Resolution Evidence → On Success	When resolution evidence is submitted	Save the evidence details and associate them with the complaint	DS-06

Table 5.3: Workflows Implemented

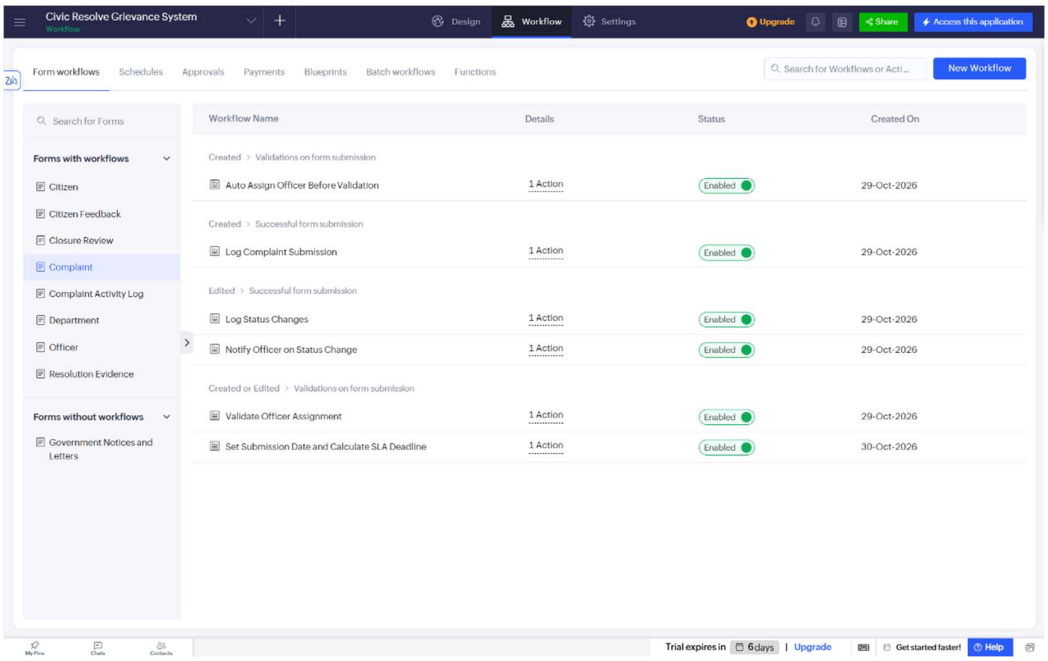


Figure 5.6: Complaint Management Workflow Configuration

5.7 Integrations

Integration	Service / Endpoint	Method	Authentication	Data Exchanged / Purpose
Internal Form Integration	Zoho Creator Forms	Built-in Lookup Fields	Zoho Creator User Authentication	Connects Citizen, Department, Officer, Complaint, Activity Log, Resolution Evidence, Closure Review, and Feedback records.
Workflow and Notification Integration	Zoho Creator Workflows and Email Notifications	Built-in Workflow Actions and Deluge	Zoho Creator User Authentication	Automates complaint updates, officer assignment, SLA calculations, and email notifications.
Application Data Management	Zoho Creator Built-in Database	Built-in Data Management	Zoho Creator User Authentication and Role-Based Permissions	Stores and manages application records across all eight forms.

Table 5.4: Integrations Implemented

Code 5.3: Internal Integration Using Zoho Creator Lookup Fields

```
// Internal integration through Zoho Creator lookup fields
// Purpose: Retrieve the related complaint record

complaintRecord = Complaint[ID == input.complaint];

if(complaintRecord.count() > 0)
{
    info "Related complaint record found";
    info complaintRecord.ID;
}
```

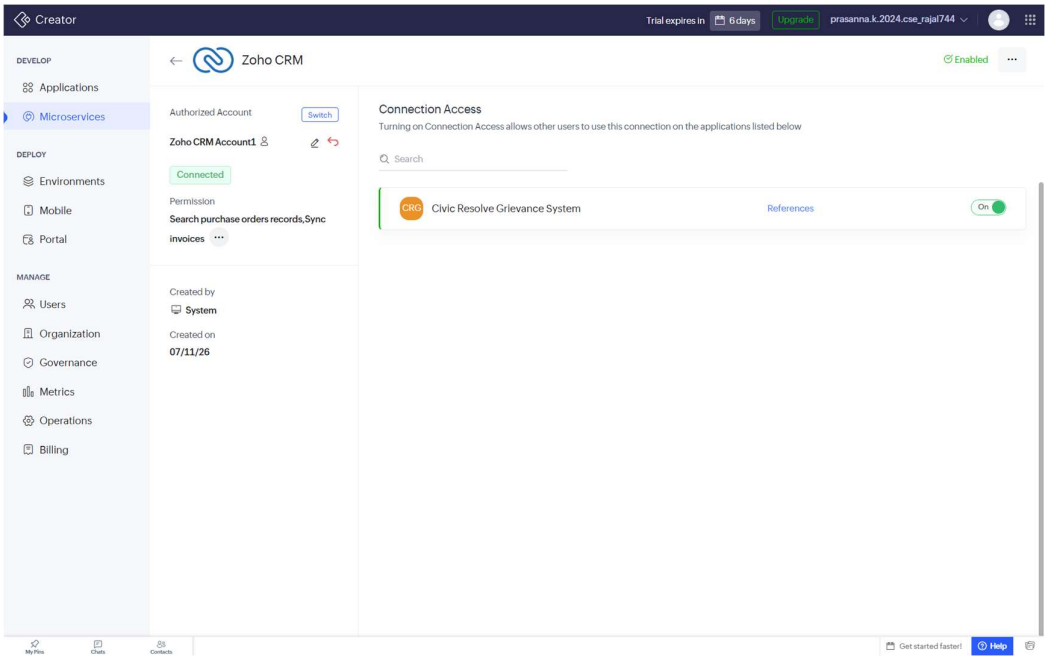


Figure 5.7: Zoho CRM Connection Setup and Authorization for Civic Resolve Grievance System

5.8 User Management and Role-Based Access

The Civic Resolve application implements role-based access control (RBAC) using Zoho Creator's user management and permission settings. Four user roles are defined: Administrator, Citizen, Officer, and Reviewer. Each role is assigned permissions according to its responsibilities within the grievance management process.

The Administrator has overall access to manage citizens, departments, officers, complaints, feedback, and application records. The Citizen role allows users to submit complaints, view their own complaint records, track progress, and provide feedback. The Officer role provides access to assigned complaints, allowing officers to update complaint status, record activities, and upload resolution evidence. The Reviewer role

allows users to examine complaint details, review submitted resolution evidence, and approve or reject closure requests.

Permission settings are configured to control Add, Edit, View, and Delete operations for the application forms and reports. Lookup relationships connect records across the eight forms, while role-based permissions help restrict access to relevant application components. Record-level access is intended to ensure that citizens can view their own complaints and officers can access complaints assigned to them. These access restrictions support data privacy and prevent unauthorized modifications.

Overall, the RBAC configuration provides controlled access to Civic Resolve and separates administrative, citizen, officer, and review responsibilities.

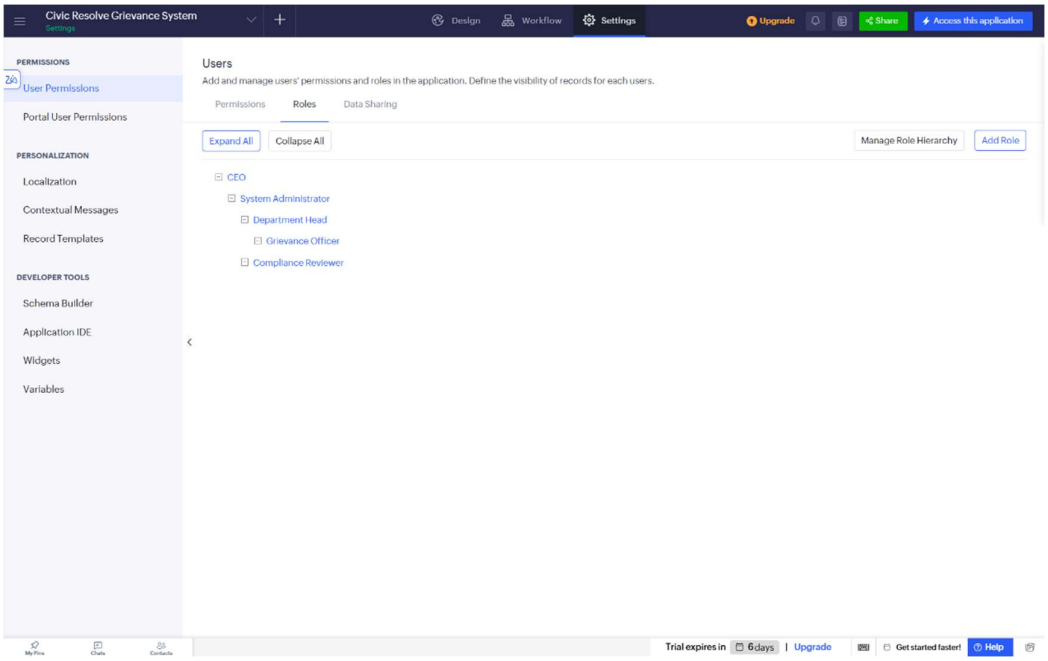


Figure 5.8: Roles and Permission Profiles Configuration of Civic Resolve

5.9 AI and Analytics Features

Civic Resolve uses Zoho Creator's built-in reports and dashboards to support data monitoring and analysis. The application provides structured complaint information that can be used to monitor total complaints, pending complaints, resolved cases, officer activities, closure review status, and citizen feedback. List reports and summary views help administrators and reviewers examine complaint records and track resolution

progress. These reports support better monitoring and decision-making by presenting application data in an organized manner.

The current implementation does not include dedicated AI features such as Zia AI prediction, sentiment analysis, OCR, keyword extraction, or automatic complaint classification. Zoho Analytics integration has also not been implemented. Deluge scripts and workflow configurations are used for rule-based automation, including officer assignment, SLA deadline calculation, complaint status updates, and notifications. These operations are based on predefined conditions rather than AI models.

AI assistance may be used during the development process to prepare or refine application logic and project documentation. Any suggested scripts are reviewed against the application's field names, tested in Zoho Creator, and corrected where necessary before use. Therefore, Civic Resolve currently focuses on workflow automation and built-in reporting rather than claiming fully implemented AI-based analytics.

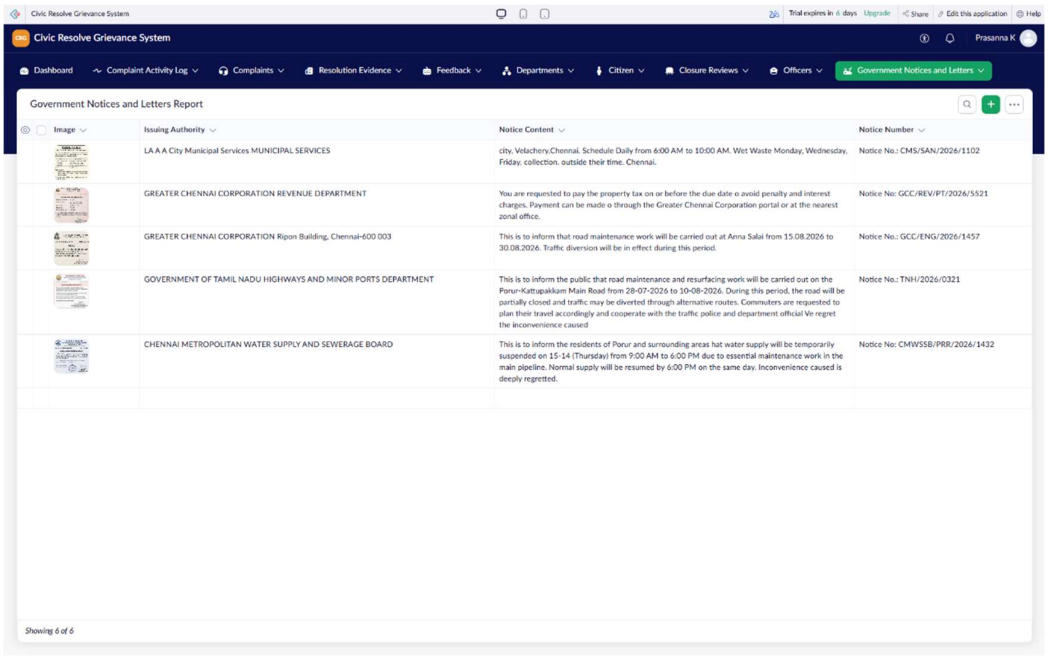


Figure 5.9: AI-Based OCR Image Processing and Text Extraction in Civic Resolve

5.10 Pro-Code Extension

The Civic Resolve application is developed entirely using the Zoho Creator low-code platform. No additional pro-code extensions such as Zoho Catalyst, JavaScript widgets, custom external applications, or Creator REST APIs have been implemented in the current version. The application's functionality is developed using Zoho Creator forms, reports, dashboards, workflows, and Deluge scripting. This approach was selected to reduce development complexity, simplify maintenance, and make use of the platform's built-in capabilities.

In future enhancements, Zoho Catalyst could be used to develop independent backend services and handle advanced processing tasks. JavaScript widgets could provide customized user interface components and interactive dashboards, while Creator REST APIs could enable secure communication with external government systems. These extensions could improve the scalability, flexibility, and integration capabilities of Civic Resolve as the project evolves.

5.11 Mobile and Portal Access

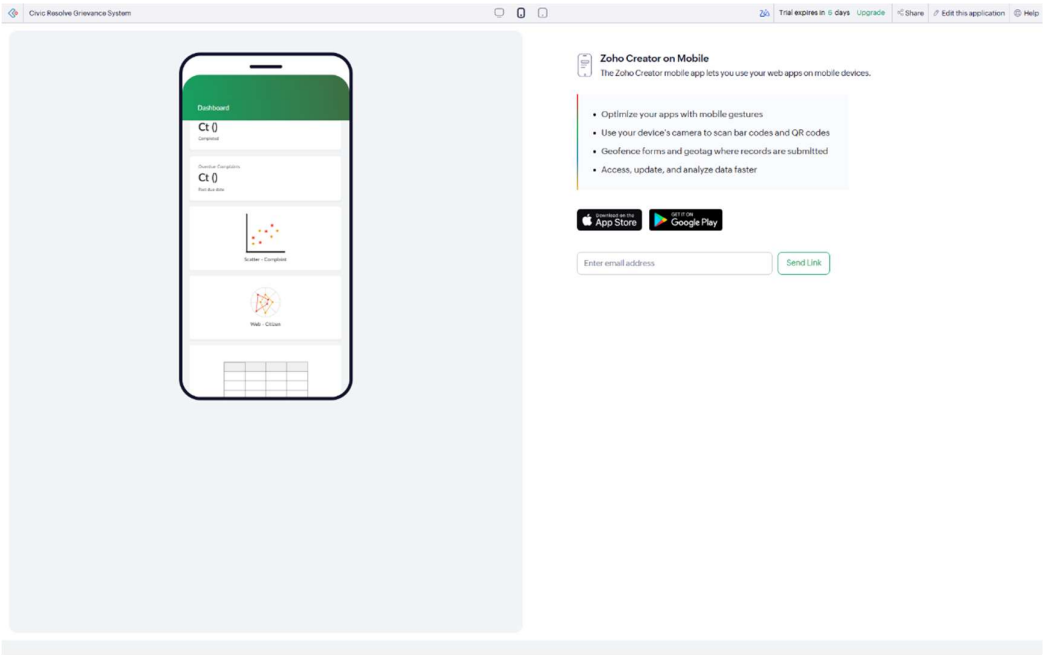


Figure 5.10: Civic Resolve Application – Mobile Interface

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

The Civic Resolve application follows a systematic testing strategy to verify the functionality, accuracy, and reliability of its forms, workflows, and role-based access controls. Unit testing is performed on individual forms to check field properties, mandatory fields, lookup relationships, and data validation. Deluge scripts are tested separately to verify operations such as officer assignment, SLA deadline calculation, complaint status updates, activity logging, and closure review processing.

Integration testing is carried out to verify communication between related forms and automated workflows. For example, when a complaint is registered, the system is checked to ensure that the appropriate officer is assigned, the SLA deadline is calculated, and the complaint details are stored correctly. Closure review testing verifies whether an approved or rejected decision updates the corresponding complaint status.

Role-based access testing checks the permissions assigned to Administrator, Citizen, Officer, and Reviewer accounts. Each role is tested to ensure that users can access the appropriate forms, reports, and records according to their permissions. User acceptance testing (UAT) focuses on the complete grievance management process, including complaint registration, assignment, status tracking, evidence submission, closure approval, and feedback collection.

Testing helps identify configuration errors, incorrect field values, workflow issues, and unauthorized access. Identified issues are corrected and the relevant functionality is tested again to ensure that Civic Resolve operates as expected.

6.2 Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-01	FR-01	Valid complaint submission	Submit completed form	Record created	Successfully created	Pass
TC-02	FR-01	Missing email	Submit without email	Validation error shown	Validation displayed	Pass
TC-03	FR-02	SLA calculation	Register complaint	Deadline calculated	Calculated successfully	Pass
TC-04	FR-03	Officer notification	Assign complaint	Notification sent	Sent successfully	Pass
TC-05	FR-04	Closure approval	Reviewer approves request	Status: Closed	Updated successfully	Pass
TC-06	FR-05	Dashboard verification	Open dashboard	Reports displayed	Displayed successfully	Pass
TC-07	FR-06	Add department/officer	Submit valid details	Records saved	Saved successfully	Pass
TC-08	FR-07	Update complaint status	status Officer updates status	Status updated	Updated successfully	Pass
TC-09	FR-08	Activity logging	Change complaint status	Activity recorded	Recorded successfully	Pass
TC-10	FR-09	Upload evidence	Upload document	Evidence saved	Saved successfully	Pass
TC-11	FR-10	Closure review	Approve closure	Complaint closed	Closed successfully	Pass
TC-12	FR-11	Citizen tracking	View own complaints	Records displayed	Displayed successfully	Pass
TC-13	FR-12	Role permissions	Access by user role	Restricted access	Worked successfully	Pass

Table 6.1: Test Cases

6.3 Debugging Deluge Scripts

Deluge scripts were debugged using info statements to display variable values, the Deluge editor's Execute option to check script execution, workflow logs to identify errors, and error messages displayed during form submission. These techniques helped identify incorrect field link names, invalid criteria, and workflow execution issues. One issue encountered was the error “Variable Status does not exist in Complaint in criteria.” It occurred because the script used Status instead of the correct field link name status. Correcting the field name resolved the issue.

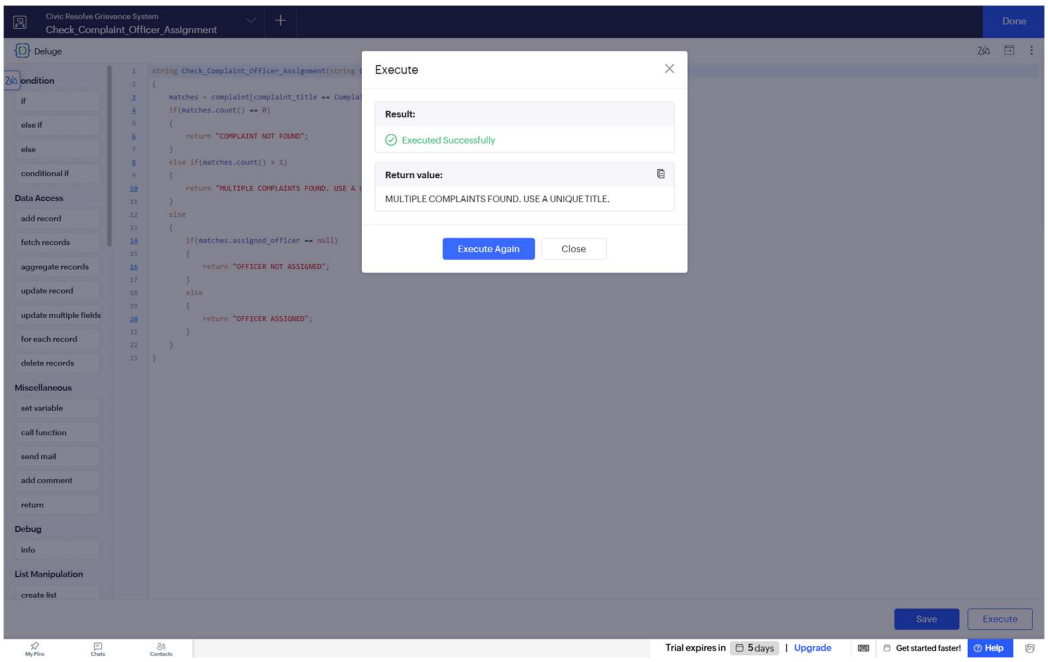


Figure 6.1: Deluge Script Editor Showing Debugging Statements

6.4 Defect Log

Bug ID	Description	Root Cause	Fix Applied	Status
BUG-01	Variable Status does not exist in Complaint criteria	Incorrect field link name (Status)	Changed Status to status	Closed
BUG-02	Closure Review workflow execution error	Incorrect record criteria or field reference	Corrected the Deluge script and verified the workflow	Closed
BUG-03	Form submission validation error	Mandatory field was missing	Entered the required field values	Closed

Table 6.2: Defect Log

6.5 User Acceptance Testing

User	Role Tested	Feedback	Action Taken
User 1	Citizen	Complaint submission and status tracking were easy to use.	No changes required.
User 2	Officer	Complaint updates and evidence submission were simple.	No changes required

Table 6.3: User Acceptance Feedback

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

The Civic Resolve application was developed and configured in the Zoho Creator Development environment, where forms, reports, dashboards, Deluge scripts, and workflows were created and tested. After resolving errors and verifying the main functionalities, the application was prepared for staging review to check form submissions, role permissions, and workflow execution. The verified application was then published through Zoho Creator for user access. Version notes were maintained to record important changes, such as workflow corrections, field updates, and permission changes. If an issue occurs after publishing, the rollback plan is to restore the previous stable application version, correct the issue in the Development environment, retest it, and publish the updated version.

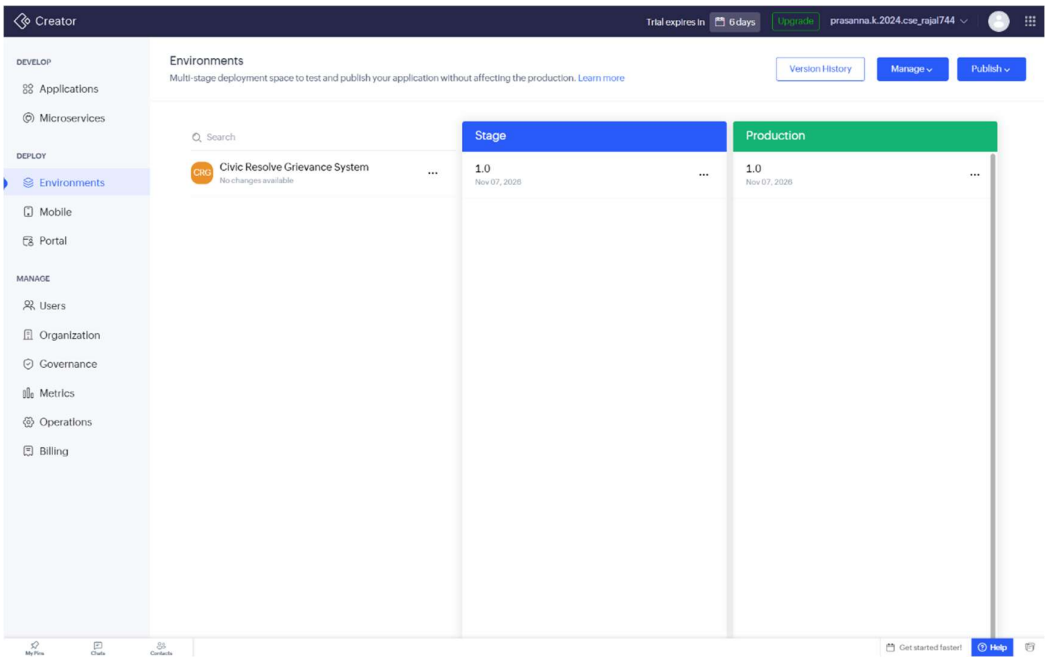


Figure 7.1: Stage and Production Environment Configuration of Civic Resolve

S.No.	Checklist Item	Done ()
1	All test cases executed in Development / Stage	✓
2	Sample/test data removed or isolated	✓
3	Roles, permissions and users configured in Production	✓
4	Connections authorised in Production	N/A

S.No.	Checklist Item	Done ()
5	Schedules and workflows enabled	✓
6	Application published and URL verified	✓
7	Version / release notes recorded	✓

Table 7.1: Deployment Checklist

7.2 Performance Optimisation

Optimisation Applied	Component	Before	After	Improvement
Replaced fetch inside loop with single criteria fetch	DS-03	Repeated database fetches	Single criteria-based fetch	Reduced database operations
Filtered report using default criteria	All Complaints Report	Displayed all complaint records	Displayed filtered records	Faster report loading
Replaced repeated fetches with a single criteria fetch	DS-03	Multiple database fetches	Single fetch	Reduced database calls
Applied default filters	All Complaints Report	All records displayed	Relevant records displayed	Reduced unnecessary data loading
Limited report columns	Complaint Reports	All fields displayed	Required fields only	Improved report readability
Used specific criteria in fetch statements	DS-01	Broad record search	Criteria-based search	Faster record retrieval
Sorted pending complaints by SLA deadline	Pending Complaints Report	Unsorted records	Deadline-based sorting	Easier identification of urgent complaints

Table 7.2: Performance Optimisation Results

7.3 Security Features

Security Control	Where Applied	Threat Mitigated
Role-based permissions	All eight forms and application pages	Unauthorised access

Security Control	Where Applied	Threat Mitigated
Record-level permission criteria	Complaint records	Unauthorised data access
Field-level hiding	Sensitive citizen and officer fields	Exposure of sensitive data
Zoho Creator authentication	Application login	Unauthorised login
Input validation	Complaint and registration forms	Invalid or incomplete data

Table 7.3: Security Controls Implemented

7.4 Governance and Administration

Civic Resolve follows basic governance and administration practices to maintain organized and secure application management. Consistent naming conventions are used for forms, fields, reports, and Deluge scripts. Zoho Creator's audit trail and application history help track administrative activities and configuration changes. Application data can be exported periodically for backup and recovery purposes. User access and permissions are managed according to the four roles: Administrator, Citizen, Officer, and Reviewer. The Administrator is responsible for monitoring application records, reviewing changes, managing users, maintaining workflows, and updating the application when required.

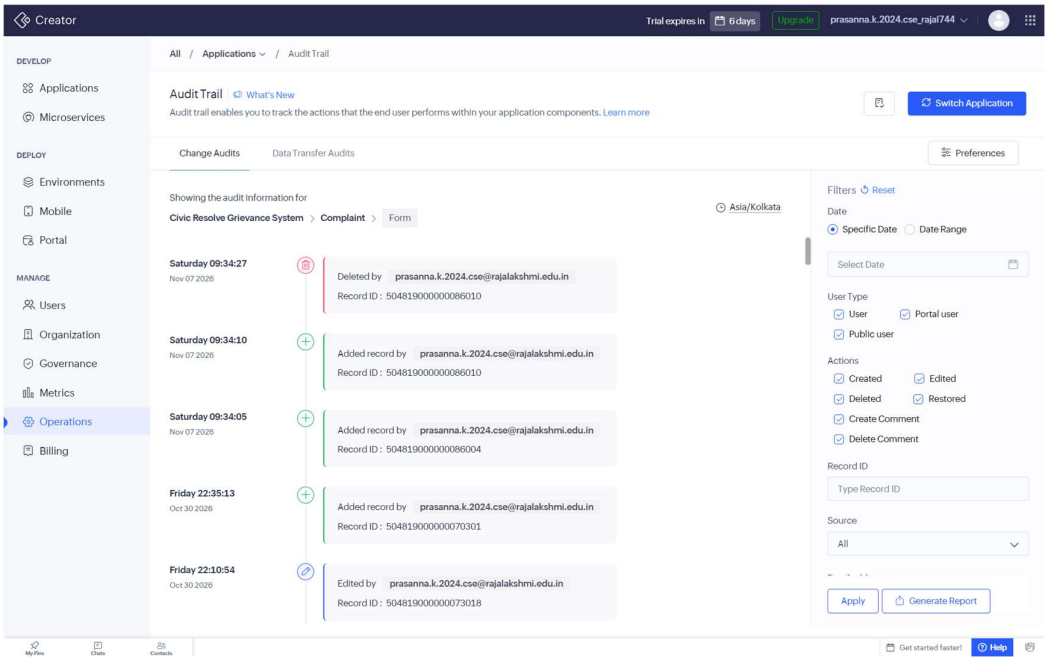


Figure 7.2: Audit Trail and Administrative Activity Monitoring of Civic Resolve

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

A citizen registers a civic complaint by entering the complaint title, description, and department details. The system saves the complaint, assigns it to an appropriate officer, and calculates the SLA deadline. The officer views the assigned complaint, updates its status, and uploads resolution evidence after completing the work. The reviewer verifies the submitted evidence and approves or rejects the closure request. Once approved, the complaint status changes to Closed. Finally, the Administrator monitors complaint records and resolution progress through reports and dashboards.

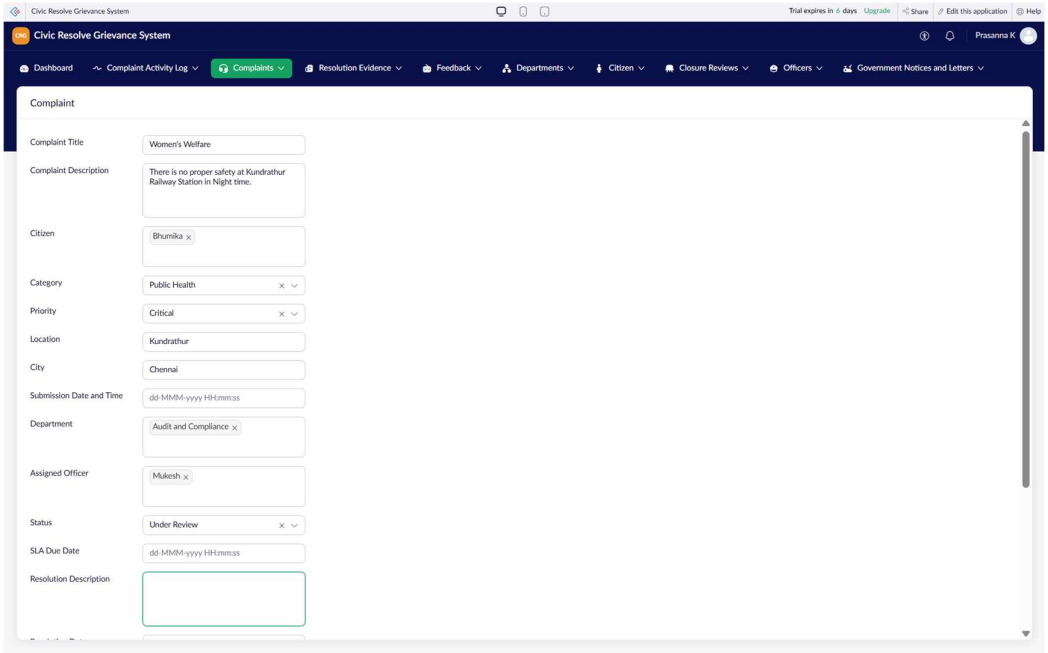


Figure 8.1: Scenario Step 1 – Citizen Complaint Registration

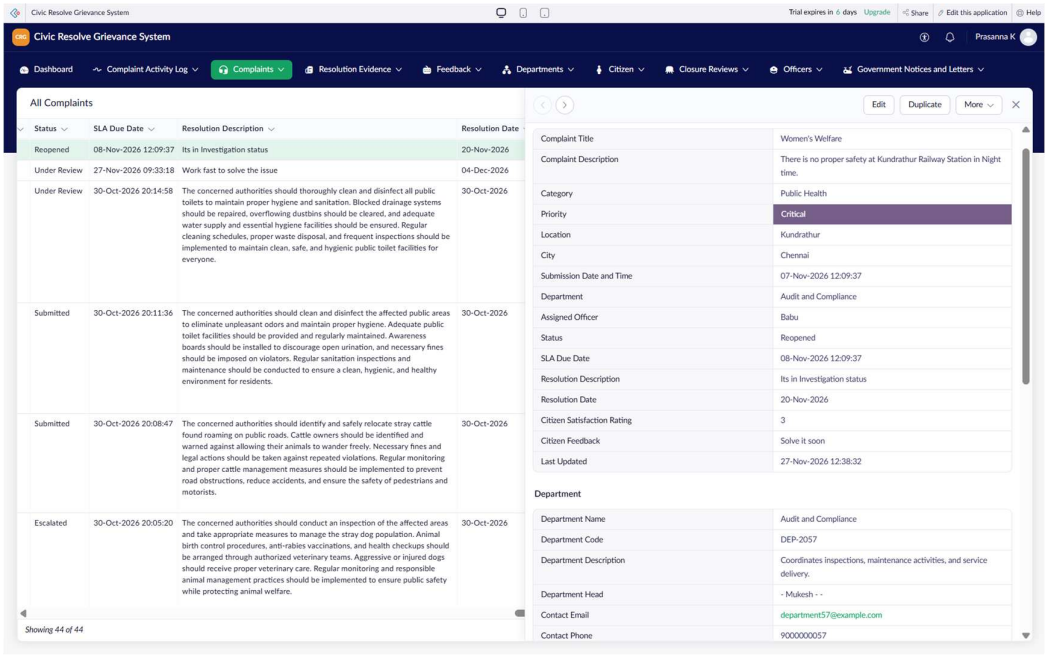


Figure 8.2: Scenario Step 2 – Officer Complaint Status Update and Evidence Submission

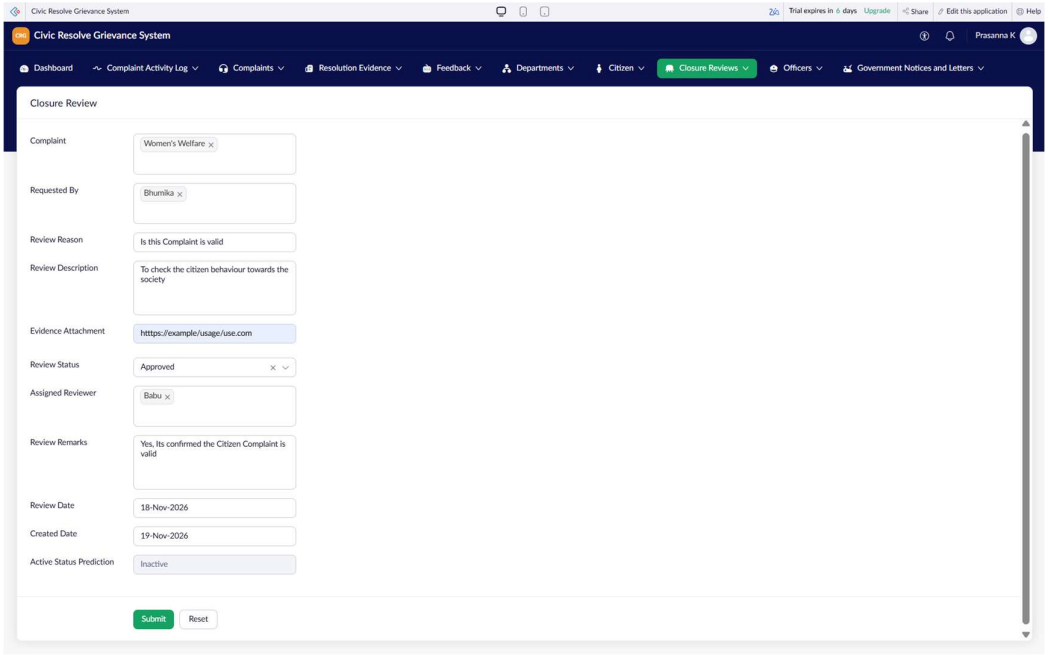


Figure 8.3: Scenario Step 3 – Complaint Closure Approval by Reviewer

8.2 Achievement of Objectives

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
Objective 1: Manage citizen, department, officer, and complaint records.	Section 5.2 – Forms and Data Capture	Eight forms created for centralized data management.
Objective 2: Automate complaint registration, assignment, SLA calculation, and closure approval.	Sections 5.5 and 5.6 – Deluge Scripts and Workflows	Automated complaint processing and status updates.
Objective 3: Monitor complaints and resolution progress.	Sections 5.3 and 5.4 – Reports and Dashboards	Reports display complaint status, pending cases, and officer activities.

Table 8.1: Mapping of Objectives to Outcomes

8.3 Comparison with the Existing System

Parameter	Existing System	Proposed Application	Improvement
Time to process one request	Manual processing	Automated workflows	Reduced processing time
Manual steps	Multiple manual steps	Automated assignment and status updates	Less manual work
Report generation	Manually prepared	Instant reports and dashboards	Faster reporting
Data accessibility	Scattered registers and spreadsheets	Centralized cloud database	Easier data access

Table 8.2: Existing System versus Proposed Application

8.4 Limitations of the Low-Code Approach

Zoho Creator provides a convenient low-code development environment, but certain limitations were encountered while developing Civic Resolve. Customising the user interface beyond the platform's built-in components was limited. Complex business logic required additional Deluge scripting and careful handling of field link names and lookup criteria. Debugging workflow errors also required repeated testing. The application was designed using standard forms, reports, dashboards, and reusable Deluge scripts to reduce these difficulties. Another limitation is vendor dependency, as the application relies on Zoho Creator's platform and services. These constraints were managed by keeping the application modular and using built-in platform features wherever possible.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

Civic Resolve uses all three levels of the No-Code → Low-Code → Pro-Code spectrum. The no-code approach was used to create forms, reports, dashboards, user roles, and application navigation through Zoho Creator's visual builder. Low-code development was used for Deluge scripts and workflows to automate officer assignment, SLA calculation, complaint status updates, activity logging, notifications, and closure approval. Pro-code development was not required in the current implementation. However, advanced features such as external government database integration, customised user interfaces, and complex AI-based complaint analysis may require technologies such as JavaScript, Python, or external APIs. I would move towards pro-code when the requirements exceed Zoho Creator's built-in capabilities or require greater flexibility, scalability, and integration.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

Civic Resolve is a centralized civic grievance management application developed using Zoho Creator. It includes eight forms for managing citizens, departments, officers, complaints, activity logs, resolution evidence, closure reviews, and feedback. The application uses Deluge scripting and workflows to automate officer assignment, SLA calculation, complaint status updates, activity tracking, notifications, and closure approval.

The project provides role-based access for Administrators, Citizens, Officers, and Reviewers. Reports and dashboards help monitor complaint progress, pending cases, and resolution activities. Overall, Civic Resolve simplifies the grievance management process by reducing manual work, improving transparency, and providing an organized platform for tracking complaints from registration to closure.

9.2 Future Enhancements

- **Predictive Complaint Analytics:** Integrate Zoho Zia to predict complaint trends and identify high-priority complaints.
- **Mobile Application:** Develop a dedicated mobile application for citizens to register complaints and track their status.
- **Advanced Integration:** Use Zoho Catalyst and external APIs to support complex processing and integration with government databases.

9.3 Learning Outcomes and Career Relevance

Developing Civic Resolve helped me gain practical experience in Zoho Creator, Deluge scripting, form design, report creation, workflow automation, and debugging. I learned how to manage application data, configure user permissions, automate business processes, and test application functionality. These skills are relevant to career roles such as Citizen Developer, Low-Code Developer, Solutions Consultant, and Automation Engineer. This project also improved my problem-solving, logical thinking, and ability to develop real-world applications using low-code technologies.

REFERENCES

[1] W. Rosam and O. Kalnichevski, Low-Code Development: A Practical Guide. New Delhi, India: BPB Publications.

[2] Zoho Corporation, “Zoho Creator Help Documentation.” [Online]. Available: https://www.zoho.com/creator/help/ (Accessed: 11 November 2026).

[3] Zoho Corporation, “Deluge Script Reference Guide.” [Online]. Available: https://www.zoho.com/deluge/ (Accessed: 11 November 2026).

[4] Zoho Corporation, “Zoho Creator Learning Centre.” [Online]. Available: https://www.zoho.com/creator/resources/ (Accessed: 11 November 2026).

[5] Gartner, Inc., “Magic Quadrant for Enterprise Low-Code Application Platforms,” Gartner Research (2023).

[6] A. C. Bock and U. Frank, “Low-Code Platform,” Business & Information Systems Engineering, vol. 63, pp. 733–740, 2021.

[7] A. Bucaioni, A. Cicchetti, and F. Ciccozzi, “Modelling in Low-Code Development: A Multi-Vocal Systematic Review,” Software and Systems Modeling, vol. 21, pp. 1959–1981, 2022.

[8] Zoho Corporation, “Understanding Audit Trail.” [Online]. Available: https://help.zoho.com/portal/en/kb/creator/developer-guide/operations/audit-trail/articles/understanding-audit-trail (Accessed: 11 November 2026).

[9] Zoho Corporation, “Export Data from Reports.” [Online]. Available: https://help.zoho.com/portal/en/kb/creator/developer-guide/reports/general-properties/articles/export-data-from-reports (Accessed: 11 November 2026).

[10] Zoho Corporation, “Configuring Deluge Script Actions for Report Workflows.” [Online]. Available: https://help.zoho.com/portal/en/kb/creator/developer-guide/workflows/report-workflows/articles/configuring-deluge-script-actions-for-report-workflows (Accessed: 11 November 2026).

[11] Zoho Corporation, “Backup and Restore Your Apps.” [Online]. Available: https://www.zoho.com/creator/newhelp/account-setup/backup-restore-apps.html (Accessed: 11 November 2026).

APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created & sections/menu organised	Y	Civic Resolve (CRG) – Application Navigation Menu	5.1	Fig. 5.1
II	Forms with ≥ 8 distinct field types	Y	Citizen, Department, Officer, Complaint and supporting forms	5.2	Fig. 5.2(a)–(h)
II	Lookup field (relationship between forms)	Y	Citizen, Department, Officer and Complaint forms	4.2, 5.2	Fig. 5.2
II	Subform (parent–child data)	Y	Complaint Form – related activity and resolution details	4.2, 5.2	Fig. 5.2
II	Field validations / layout rules	Y	Citizen, Department, Officer and Complaint forms	5.2	Fig. 5.2
II	List report with filter, grouping, custom action	Y*	All Complaints Report	5.3	Fig. 5.4
II	Kanban / Calendar / Timeline / Map report	Y	Complaint status and tracking reports	5.3	Fig. 5.3
II	Pivot table / chart report	N	Not implemented	5.3	—
II	Page / dashboard built with UI Builder	Y	Dashboard – Administrator Dashboard of Civic Resolve	5.4	Fig. 5.5
II	Deluge – form events (validate / success / field actions)	Y	Complaint Form – On Validate, On Success and field actions	5.5	Fig. 5.6
II	Deluge – functions, fetch, loops, collections	Y*	Complaint assignment, SLA calculation and activity tracking	5.5	Code 5.5.1–5.5.3

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
II	Email / in-app notifications	Y	Complaint notification and status update workflows	5.5, 5.6	Fig. 5.6
II	Form workflow	Y	Complaint Form – AI Complaint Analysis on Submit and Update	5.6	Fig. 5.6
II	Scheduled workflow	Y	Scheduled SLA monitoring and overdue complaint tracking	5.6	Fig. 5.6
II	Approval process	Y	Complaint Closure Review – Approval / Rejection	5.6	Fig. 5.6
II	Blueprint	Y	Complaint Management – Complaint Status Transition Process	5.6	Fig. 5.6
II	Batch / report workflow	Y	Central Complaint Intelligence Engine, Duplicate Detection Engine, Department Performance Optimizer and Executive Intelligence Dashboard	5.6	Fig. 5.6
II	Testing & debugging evidence	Y	Application testing, workflow execution and debugging	6	Fig. 6.1–6.3
III	Integration with a Zoho service	Y*	Civic Resolve – Zoho service integration	5.7	Fig. 5.7
III	Connection / REST API call (invokeurl)	N	Not implemented	5.7	—
III	Webhook or Creator REST API	N	Not implemented	5.7	—
III	Users, roles and permission profiles	Y	User Management – System Administrator, Department Head	5.8	Fig. 5.8

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
III	Record-level access criteria / customer portal	Y*	Civic Resolve – Public Portal; portal permissions and citizen-specific record access (after configuration and testing)	5.8	Fig. 5.8
III	Zia / AI feature	Y	Complaint Form – AI Complaint Analysis and six AI fields	5.9	Fig. 5.9
III	Analytics (charts / Zoho Analytics)	N	Not implemented	5.9	—
IV	Performance optimisation applied & measured	N	Not implemented	7.2	—
IV	Deployment across environments	N	Development environment only	7.1	—
IV	Security controls (encryption, OAuth, permissions)	Y	Application Security and User Permission Settings	7.3	Fig. 7.3
IV	Governance (audit trail, administration)	Y	Application Administration and Audit Trail	7.4	Fig. 7.4
V	Pro-code extension (Catalyst / widgets / custom functions)	N	Not implemented	5.10	—
V	Low-code limitations & spectrum analysis	Y*	Technical Analysis – Low-Code Limitations and Spectrum Analysis	8.4, 8.5	Section 8.4–8.5
—	Mobile app / portal access	Y	Civic Resolve – Portal Login and Dashboard Access (after testing)	5.11	Fig. 5.11

Table A.1: Feature Coverage Checklist

Total features implemented: 25 / 32

APPENDIX B – COURSE OUTCOME MAPPING

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2	3	
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6	3	
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7–5.9	2	
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7	2	
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3	3	

Table B.1: Mapping of Course Outcomes to Report Chapters

APPENDIX C – COMPLETE DELUGE SOURCE CODE

C.1 Automatic Officer Assignment

```
// Script ID : DS-01
// Location  : Complaint Form → On Success
// Purpose   : Automatically assign an officer to a newly submitted
complaint.

if(input.department != null)
{
	for each  deptId in input.department
	{
		dept = department[ID == deptId];
		if(dept.count() > 0 && dept.department_head != null)
		{
			for each  officerId in dept.department_head
			{
				head = officer[ID == officerId];
				if(head.count() > 0 && head.active_status ==
"Active")
				{
					input.assigned_officer = head.ID;
					break;
				}
			}
		}
		if(input.assigned_officer != null)
		{
			break;
		}
	}
}
```

C.2 SLA Deadline Calculation

```
// Script ID : DS-02
// Location  : Complaint Form → On Success
// Purpose   : Calculate and update the complaint's SLA deadline.

// Automatically set submission date and time
if(input.submission_date_time == null)
{
	input.submission_date_time = zoho.currenttime;
}
// Automatically calculate SLA due date if not entered
if(input.sla_due_date == null)
{
	input.sla_due_date = input.submission_date_time.addDay(1);
}
```

C.3 Complaint Status Update and Activity Logging

```
// Script ID : DS-03
// Location  : Complaint Form → On Update
// Purpose   : Update complaint status and maintain its activity log.

// Get previous activity records
logs = complaint_activity_log[complaint == input.ID];
latestLogID = 0;
previousStatus = "Initial";
previousOfficer = null;
previousDepartment = null;
for each  logRecord in logs
{
      if(logRecord.ID > latestLogID)
      {
            latestLogID = logRecord.ID;
            previousStatus = logRecord.new_status;
            previousOfficer = logRecord.new_officer;
            previousDepartment = logRecord.new_department;
      }
}
// Identify the type of update
actionName = "Complaint Details Updated";
if(input.status != previousStatus)
{
      actionName = "Status Changed";
}
// Save complete complaint history
insert into complaint_activity_log
[
      complaint=input.ID
      previous_status=previousStatus
      new_status=input.status
      previous_officer=previousOfficer
      new_officer=input.assigned_officer
      previous_department=previousDepartment
      new_department=input.department
      action_performed=actionName
      changed_by=zoho.loginuser
      activity_date_time=zoho.currenttime
      Added_User=zoho.loginuser
]
```

C.4 Complaint Notification

```
// Script ID : DS-04
// Location  : Complaint Form → Relevant Workflow
// Purpose   : Send notifications when a complaint is assigned or its
status changes.

if(input.assigned_officer != null)
{
      assigned = officer[ID == input.assigned_officer];
      if(assigned.count() > 0 && assigned.email != null &&
assigned.email != "")
      {
            sendmail
            [
                  from :zoho.adminuserid
                  to :assigned.email
                  subject :"Complaint Status Update"
                  message :"Complaint ID: " + input.ID + "\nCurrent
Status: " + input.status
            ]
      }
}
```

C.5 Resolution Evidence Management

```
// Script ID : DS-05
// Location  : Resolution Evidence Form → On Success
// Purpose   : Associate uploaded resolution evidence with the
corresponding complaint.

oldStatus = old.verification_status;
insert into complaint_activity_log
[
      complaint=input.complaint
      previous_status=oldStatus
      new_status=input.verification_status
      action_performed="Evidence Verification Updated"
      changed_by=zoho.loginuser
      activity_date_time=zoho.currenttime
      Added_User=zoho.loginuser
]
```

C.6 Closure Review Processing

```
// Script ID : DS-06
// Location  : Closure Review Form → On Success
// Purpose   : Process reviewer decisions and update the linked
complaint status.

if(input.complaint_selection != null)
{
	complaint_id = input.complaint_selection.toLong();
	if(input.review_status == "Approved")
	{
		try
		{
			// Move the Blueprint stage to "Closed"
			thisapp.blueprint.changeStage("complaint",
"complaint_lifecycle_management","Closed",complaint_id);
		}		catch (e)
		{
			// Fallback direct update if Blueprint is inactive
			update complaint[ID == complaint_id]
			[
				status="Closed"
			];
		}
	}
	else if(input.review_status == "Rejected")
	{
		try
		{
			// Move the Blueprint stage to "Under Review"
			thisapp.blueprint.changeStage("complaint",
"complaint_lifecycle_management","Under Review",complaint_id);
		}		catch (e)
		{
			// Fallback direct update if Blueprint is inactive
			update complaint[ID == complaint_id]
			[
				status="Under Review"
			];
		}
	}
}
```

APPENDIX D – USER MANUAL

D.1 Accessing the Application

1. Open Google Chrome or any supported web browser.
2. Navigate to the Civic Resolve application URL.
3. Enter your registered Zoho account credentials.
4. Log in to access the application.
5. The available forms, reports, and dashboards depend on the user's assigned role.
6. Mobile users can access the application through the Zoho Creator mobile application by logging in with their registered account.

D.2 Instructions for Citizen

- Log in to the Civic Resolve application.
- Open the Complaint form.
- Enter the complaint title, description, and required details.
- Select the appropriate department.
- Submit the complaint.
- Open the complaints report to view submitted complaint details and track progress.
- Open the Feedback form to submit feedback after resolution.

D.3 Instructions for Grievance Officer

- Log in using your registered account.
- Open the assigned complaints report.
- Select the complaint assigned to you.
- Review the complaint details and investigate the issue..
- Update the complaint status and enter remarks.
- Upload supporting documents or images through the Resolution Evidence form.
- Submit a closure request when the issue is resolved.

D.4 Instructions for Department Head

- Log in to Civic Resolve.
- Open the department-related reports.
- Review complaints assigned to the department.
- Monitor officer assignments and complaint progress.
- Check pending and resolved complaints.
- Use the dashboard to monitor departmental performance.

D.5 Instructions for Compliance Reviewer

- Log in using your registered account.
- Open the Closure Review form or report.
- View pending complaint closure requests.
- Select a complaint and review the submitted resolution evidence.
- Enter review remarks.
- Select Approved or Rejected as appropriate.
- Submit the review decision.
- Verify the updated complaint status.

D.6 Instructions for System Administrator

- Log in using the administrator account.
- Open the application dashboard.
- Manage citizen, department, and officer records.
- Monitor all complaint records and assignments.
- Review complaint activity logs and resolution evidence.
- Monitor closure reviews and citizen feedback.
- Access reports and dashboards to analyse complaint progress.
- Manage user roles, permissions, and application configurations.

D.7 Accessing Reports and Dashboards

- Log in to Civic Resolve.
- Navigate to the Reports section.
- Select the required report, such as All Complaints, Officer Activities, Closure Reviews, or Feedback.
- Apply available filters to view specific records.
- Open the dashboard to monitor complaint statistics and resolution progress.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	Prasanna.K	Register No.	2116240701396
Project Title	Civic Resolve – AI Powered Grievance Management System	Date	11-11-2026

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Mark Allotted
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		**100**	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner

Name: _____

Signature of External Examiner

Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	Prasanna. K	Register No.	2116240701396
Department / Section	Computer Science and Engineering	Date of Viva	11-11-2026
Application URL Verified	Yes / No	Demo Duration	_____ min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	**100**	

Examiners' Remarks:

_____

_____

INTERNAL EXAMINER

Name: _____

EXTERNAL EXAMINER

Name: _____